

torch.pca_lowrank#

https://pytorch.org/docs/stable/generated/torch.pca_lowrank.html#torch.pca_lowrank

Description:

Performs linear Principal Component Analysis (PCA) on a low-rank matrix, batches of such matrices, or sparse matrix. This function returns a namedtuple (U, S, V) which is the nearly optimal approximation of a singular value decomposition of a centered matrix A such that $A \approx U \text{diag}(S) V^H$. The relation of (U, S, V) to PCA is as follows: A is a data matrix with m samples and n features

Function Signature

```
torch.pca_lowrank(A, q=None, center=True, niter=2)[source]#
```

Parameters

Return type

Returns:

tuple[torch.Tensor, torch.Tensor, torch.Tensor]

Code Examples

- Nathan Halko, Per-Gunnar Martinsson, and Joel Tropp, Finding structure with randomness: probabilistic algorithms for constructing approximate matrix decompositions, arXiv:0909.4061 [math.NA; math.PR], 2009 (available at ``arXiv <http://arxiv.org/abs/0909.4061>`_`).

Notes & Warnings

NOTE

Note The relation of (U, S, V) to PCA is as follows: AAA is a data matrix with m samples and n features the VVV columns represent the principal directions $S_{**2}/(m-1)$ $S^{**2} / (m - 1)$ $S_{**2}/(m-1)$ contains the eigenvalues of $ATA/(m-1)$ $A^T A / (m - 1)$ $ATA/(m-1)$ which is the covariance of A when center=True is provided. `matmul(A, V[:, :k])` projects data to the first k principal components

NOTE

Note Different from the standard SVD, the size of returned matrices depend on the specified rank and q values as follows: UUU is m x q matrix SSS is q-vector VVV is n x q matrix

NOTE

Note To obtain repeatable results, reset the seed for the pseudorandom number generator

Detailed Documentation

Documentation

Performs linear Principal Component Analysis (PCA) on a low-rank matrix, batches of such matrices, or sparse matrix.

This function returns a namedtuple (U, S, V) which is the nearly optimal approximation of a singular value decomposition of a centered matrix AAA such that $A \approx U \text{diag}(S) V^H$

The relation of (U, S, V) to PCA is as follows:

AAA is a data matrix with m samples and n features

the V columns represent the principal directions

$S_{**2}/(m-1)$ contains the eigenvalues of $A^T A / (m - 1)$ which is the covariance of A when center=True is provided.

`matmul(A, V[:, :k])` projects data to the first k principal components

Different from the standard SVD, the size of returned matrices depend on the specified rank and q values as follows:

To obtain repeatable results, reset the seed for the pseudorandom number generator

A (Tensor) – the input tensor of size $(*, m, n)$

q (int, optional) – a slightly overestimated rank of AAA. By default, $q = \min(6, m, n)$.

center (bool, optional) – if True, center the input tensor, otherwise, assume that the input is centered.

niter (int, optional) – the number of subspace iterations to conduct; niter must be a nonnegative integer, and defaults to 2.

tuple[torch.Tensor, torch.Tensor, torch.Tensor]
